

Sol ut i on Requi r ement s

Date	03-07-2026
TeamID	
Proj ect Name	Smart Agri cul tural Product i on Opt i mi zat i on Engi ne
Maxi mumMarks	4 Marks

Defi ne Funct i onal and Non- Funct i onal Requi r ement s:

Create a comprehensi ve l i st of requi r ement s t o def i ne exact l y what your sol ut i on wi l l do and how wel l i t wi l l perform

- Funct i onal Requi r ement s (FR) def i ne speci fi c behavi ors, funct i ons, or feat ures of t he syst em(What t he syst em shoul d do).
- Non- Funct i onal Requi r ement s (NFR) speci fy t he cri t eri a t hat j udge t he operat i on of a syst em, rat her t han speci fi c behavi ors (How t he syst em shoul d be).

Fi l l out t he descri pt i ons and necessary det ai l s for each cat egory provi ded bel ow.

St ep 1: Funct i onal Requi r ement s (FR)

S.No	Requi r ement Cat egory	Requi r ement Descri pt i on	Pri ori ty (Hi gh/ Medi um/ Low)
1	Aut hent i cat i on	Farmer shal l securel y l og i n usi ng Mobi l e Number + OTP veri fi cat i on. Syst em shal l val i dat e farmer I Dagai nst regi st ered dat abase bef ore al l owi ng access t o crop suggest i ons.	Hi gh

2	Aut hori zat i on l evel s	Def i ne 2 user r ol es: 1 Farmer: Can i nput farm dat a, vi ew crop suggest i ons, access market pri ces. 2 Admi n/ Agri Off i cer: Can manage ML model s, updat e Govt DB, vi ew al l farmer anal yt i cs, generat e report s.	Hi gh
3	Ext ernal i nt er faces	Syst em shal l i nt egrat e wi th: Weat her API - For real -t i me rai nfal l , t emperat ure, humi di ty dat a based on GPS Locat i on Govt agri DB - For fet chi ng dai ly market pri ces of crops from l ocal mandi s SMS Gat eway - To send crop suggest i ons vi a SMS t o farmer s wi t hout smart phones	Hi gh
4	Transact i ons processi ng	When farmer submi t s dat a, syst em shal l process i nput s t hrough ML model wi t hi n 5 seconds and ret urn t op 3 crop suggest i ons. Syst em shal l l og every t ransact i on: Farmer I D, Ti mest amp, I nput s, Suggest i ons gi ven, i nt o D1: Farmer DB. No payment processi ng requi red - servi ce i s free for farmers.	Hi gh

5	Reporting	System shall generate: Farmer Report: Individual crop history, profit/loss tracking in PDF format Admin Dashboard: District-wise crop trends, most suggested crops, ML model accuracy stats Data Export: Admin can export farmer data to Excel / CSV for Govt analysis	Medium
6	Business rules, etc.	Crop Season Rule: System shall suggest only crops suitable for current season based on date and rainfall data. Location Constraint: Suggestions must be limited to crops viable for the farmer's soil type and district. Minimum Farm Size: If farm size < 0.5 acre, system shall display warning: "Commercial profit may be low". Price Threshold: Suggest crops only if market price > minimum support price (MSP) from Govt DB.	High
7	Compliance to laws or regulations	Data Privacy: Farmer personal data (Mobile, Location) shall be encrypted and stored as per IT Act 2000 & DPDP Act 2023, India. Consent: System must take explicit consent before accessing GPS location. Agri Data Standards: Market price data fetched from Govt	High

		sources shall comply with Digital India open data policies.	
8	Other	Multilingual Support: System shall provide crop suggestions in Telugu, Hindi, and English. Offline SMS: For feature phones, system shall send crop names + profit estimate via SMS in local language. Feedback Loop: Farmer can rate suggestion accuracy after harvest. System shall use feedback to retrain ML model.	Medium
S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

Step 2: Non-Functional Requirements (NFR)

1	Performance & Speed	System shall generate crop suggestions within 3 seconds after farmer submits data. Weather API and Market Price DB calls shall complete within 2 seconds. Mobile app home page shall load in < 2 seconds on 3G network.	Response time < 3 sec
2	Scalability	System shall support 50,000 concurrent farmers during peak sowing season. Database D1 shall handle 10 lakh+ farmer records without performance drop. ML model inference shall auto-scale on	50,000 concurrent users

		cloud during high traffic.	
3	Security & Data Privacy	All farmer data (Mobile, GPS, Soil data) shall be encrypted using AES-256 at rest and TLS 1.3 in transit. OTP authentication shall expire in 5 minutes to prevent misuse. System shall conduct quarterly vulnerability scans and comply with DPDP Act 2023, India.	End-to-end AES-256 encryption
4	Reliability & Availability	System shall maintain 99.5% uptime per month, excluding scheduled maintenance. If Weather API fails, system shall use cached rainfall data from last 7 days and show warning to user. Daily automated backups of D1: Farmer DB shall be stored for 30 days.	99.5% uptime per month
5	Usability & Accessibility	User Experience: Interface shall use large icons + minimal text for farmers with low digital literacy. Complete task in ≤ 3 taps. Language: App shall support Telugu, Hindi, English with voice input/output for illiterate farmers. Accessibility: Comply with WCAG 2.1 AA standards - high contrast mode, screen reader support, font size adjustment. Offline Help:	WCAG 2.1 AA, ≤ 3 taps

		SMS version shall work on basic keypad phones without internet.	
6	Other	Maintainability: ML model shall be retrained monthly using new farmer feedback data. Code shall be modular for easy updates. Portability: Backend shall be cloud-agnostic - deployable on AWS/ Azure/ GCP. Mobile app works on Android 8.0+. Environmental: System shall promote sustainable farming by prioritizing low-water crops in drought-prone areas. Cost: Operational cost shall be < ₹2 per farmer per month for Govt to sponsor.	Cloud-agnostic, < ₹2 / farmer